FOUNDRY NOTES

Moulds, Moulding Sands and Sand-Mould Preparation

Reference framework: S. K. Hajra Choudhury & A. K. Hajra Choudhury, Elements of Workshop Technology, Vol. I

1. Mould and Moulding - Definitions and Basic Terms

1.1 Mould

Definition: A mould is a hollow cavity of the required shape prepared in a suitable moulding material. Molten metal is poured into this cavity and allowed to solidify; after solidification, the metal piece obtained has substantially the shape of the cavity.

- **Purpose:** To provide the negative impression of the required casting and to contain molten metal until it solidifies.
- **Main parts:** For a conventional sand mould, the upper half is called the cope and the lower half the drag. An additional intermediate part, when used, is called a cheek.
- **Cavity formation:** The mould cavity is generally formed by withdrawing a pattern from compacted moulding material.
- **Associated passages:** A practical mould also contains a gating system for metal entry and often risers/feeders for supplying liquid metal during solidification.

1.2 Moulding

Definition: Moulding is the operation of preparing the mould cavity in a suitable moulding material around a pattern, together with the necessary gates, runners, risers, vents and other features required for producing a sound casting.

> **Important distinction:** A pattern is the replica used to create the cavity; the mould is the prepared cavity into which molten metal is poured; the casting is the solid metal component obtained after solidification.

Term	Meaning
Pattern	A model or replica of the component used to produce the mould cavity.
Mould	The prepared cavity that receives molten metal.
Core	A separately prepared sand body placed inside the mould to form internal holes, passages or recesses.
Cope	Upper half of a two-part moulding flask.
Drag	Lower half of a two-part moulding flask.
Parting line	Plane or surface along which the mould separates into cope and drag.
Flask	Rigid frame that supports and contains the moulding sand.

2. Types of Moulds

Moulds can be classified in more than one way. For foundry study, the most useful classification is according to the mould material and whether the mould is destroyed after one casting.

2.1 Expendable or Temporary Moulds

These moulds are broken or otherwise destroyed to remove the casting. They are therefore prepared again for each casting.

- **Green-sand mould:** Made from moist sand containing clay and water. It is used without drying and is the most common sand mould for general foundry work.
- **Dry-sand mould:** A green-sand mould that is dried or baked before pouring. Drying increases strength and reduces moisture-related gas defects.
- **Skin-dried mould:** Only the surface layer adjacent to the mould cavity is dried, while the bulk remains green. It combines some advantages of green and dry moulding.
- **Loam mould:** Made from a loam mixture containing a comparatively high proportion of clay. It is generally used for very large castings and moulds may be built directly to shape using sweeps or skeleton patterns.
- **CO2 / sodium-silicate mould:** Prepared with sand bonded by sodium silicate and hardened by passing carbon dioxide gas. It develops high strength rapidly.
- **Shell mould:** A thin shell of resin-coated sand is formed around a heated metal pattern. It gives good surface finish and dimensional accuracy.
- **Cement-bonded or other chemically bonded moulds:** Used where greater strength or special mould characteristics are required.

2.2 Permanent Moulds

Permanent moulds are made of durable materials such as cast iron, steel, graphite or refractory materials and can be reused for many castings. They are common in gravity die casting and related processes.

- **Advantages:** High production rate, better dimensional consistency and generally improved surface finish.
- **Limitations:** Higher mould cost, restrictions on shape and alloy, and possible difficulty in producing intricate internal cavities.

Basis	Temporary / Expendable Mould	Permanent Mould
Reusability	Normally one casting per mould	Used repeatedly
Typical material	Sand or other expendable refractory material	Metal, graphite or durable refractory
Tooling cost	Lower mould cost	Higher initial mould cost
Complex shapes	Very suitable, especially with cores	More limited
Production rate	Low to medium	Medium to high
Examples	Green sand, dry sand, shell, CO2 sand	Gravity die / permanent-mould casting

3. Moulding Sand - Functions and Requirements

Moulding sand is the principal refractory material used for making sand moulds and cores. It must withstand contact with molten metal, reproduce the pattern shape, permit gas escape and finally allow the casting to contract and be removed without excessive difficulty.

- **Base sand:** Provides the refractory granular framework.
- **Binder:** Binds individual grains together and develops green, dry or chemical strength.
- **Moisture or hardener:** Activates certain binders or causes chemical hardening.
- **Additives:** Modify properties such as surface finish, collapsibility, resistance to metal penetration and hot strength.

4. Properties of Moulding Sands

A good moulding sand must possess a balanced combination of properties. Improving one property may sometimes reduce another, so the sand mixture is selected according to the metal, casting size, section thickness and moulding method.

Refractoriness: Ability of the sand to withstand the high temperature of molten metal without softening, fusing or reacting excessively. It depends mainly on the purity and mineral composition of the base sand. High refractoriness is especially necessary for steel and heavy castings.

Permeability (porosity): Ability of the compacted mould to permit air, steam and gases to pass through it. Adequate permeability prevents gas pressure, blows and pinholes. It depends on grain size, grain shape, clay content, moisture and degree of ramming.

Strength / cohesiveness: Ability of sand grains to remain bonded so that the mould retains its shape during pattern withdrawal, handling and pouring. Strength is considered as green strength, dry strength and hot strength depending on the condition.

Green strength: Strength of the sand in the moist, undried condition. It is essential for handling green-sand moulds and resisting erosion during pouring.

Dry strength: Strength after moisture has been removed. It is important in dry moulds and for surfaces exposed to prolonged heating.

Hot strength: Strength of the moulding sand at elevated temperature. Sufficient hot strength is required to resist erosion, cracking and metal pressure during pouring and early solidification.

Plasticity / flowability: Ability of the sand mixture to flow, pack and reproduce the details of the pattern under ramming or squeezing. Good flowability promotes uniform mould density.

Adhesiveness: Ability of moulding sand to adhere to the walls of the moulding box or flask, helping the mould remain supported during handling.

Collapsibility: Ability of the sand to break down after the metal has solidified. Good collapsibility permits casting contraction and makes shakeout easier, reducing hot tears and residual stress.

Fineness: Related to grain size and distribution. Fine sand gives better surface finish, whereas coarser sand generally gives greater permeability.

Durability / reusability: Ability of the sand to retain useful properties after repeated contact with molten metal and reclamation. Good durability reduces sand consumption and waste.

Thermal stability: Ability to resist excessive expansion, cracking, scabbing or distortion when rapidly heated by molten metal.

Chemical inertness: Ability to resist undesirable chemical reaction with molten metal and slag, which could cause burn-on, inclusions or poor casting surface.

Ease of preparation and control: The mixture should respond predictably to mulling, moisture control, additives and mechanical handling in the foundry.

> **Exam point:** Permeability and strength must be balanced. Very high clay content or excessive ramming may improve strength but reduce gas permeability. Similarly, very fine grains improve surface finish but can lower permeability.

5. Types of Moulding Sands

The terms below describe moulding sands according to their condition, location in the mould or specific function.

Type of sand	Main use / characteristic
Green sand	Moist sand used directly for moulding without drying. It normally contains silica sand, clay/bentonite, water and small quantities of additives.
Dry sand	Sand mould that has been dried or baked after preparation. It has better dry strength and is suited to larger or heavier castings.
Loam sand	A high-clay sand mixture, traditionally used in moulding very large castings. The mould is shaped, coated and dried before pouring.
Facing sand	Freshly prepared, high-quality sand placed directly against the pattern and therefore in contact with molten metal. It requires high refractoriness and good surface characteristics.

Type of sand	Main use / characteristic
Backing sand / floor sand	Sand placed behind the facing sand to fill the remainder of the flask. It often contains reclaimed or used sand.
System sand	In mechanized foundries the entire mould may be made from one controlled sand mixture called system sand; separate facing and backing sands are then unnecessary.
Parting sand	Dry, fine sand or parting material used on parting surfaces and pattern surfaces to prevent sticking between cope and drag or between pattern and moulding sand.
Core sand	Special sand used for cores. It normally requires good strength, permeability, refractoriness and collapsibility, and may use oil, resin, sodium silicate or other binders.
Oil sand	Sand bonded with organic oil-type binders, traditionally used mainly in core making.
Synthetic sand	Prepared by mixing clean base sand with a controlled amount of binder and additives rather than using naturally bonded sand as dug.
Natural sand	Sand obtained from natural deposits and containing naturally occurring clay that can provide bonding after moisture is added.

6. Green Sand

Definition: Green sand is a moist moulding mixture used in the undried condition. The word “green” refers to the presence of moisture and the mould being used without baking; it does not refer to colour.

6.1 Why green sand is widely used

- Low cost and easy preparation.
- Good productivity because no separate drying operation is required.
- Suitable for hand moulding as well as high-production mechanized moulding.
- Sand can be reclaimed and reused after proper conditioning.
- Good combination of mouldability, collapsibility and general-purpose performance when properly controlled.

6.2 Limitations of green-sand moulding

- Moisture produces steam and gases when contacted by molten metal; poor permeability can therefore cause blowholes or pinholes.
- Green moulds are weaker than fully dried or chemically hardened moulds.
- Poor moisture control may cause erosion, swelling, scabs or dimensional variation.
- Very heavy castings or alloys poured at very high temperature may require special sand systems, coatings or stronger moulding methods.

7. Composition of Green Sand

A green-sand mixture is not simply wet silica sand. It is an engineered combination of base sand, clay/bentonite binder, water and selected additives. Exact percentages vary with foundry practice, metal and sand system; therefore typical ranges are used rather than one universal composition.

Constituent	Typical role in green sand	Typical proportion / note
Silica sand / base sand	Provides the refractory skeleton and controls grain size, permeability and surface finish.	Largest fraction; commonly around 80-95% of the total mixture depending on binder/additive level.
Clay or bentonite	Acts as the principal inorganic binder. In the presence of water it coats and bonds the sand grains.	Often several percent of the mixture; commonly about 5-10% in many practical green-sand systems.
Water / moisture	Activates clay bonding and gives plasticity and green strength.	Usually a few percent; controlled carefully because too little gives weak moulds and too much lowers permeability and can cause gas defects.
Carbonaceous additives such as sea	Improve surface finish and reduce metal penetration/burn-on by forming a reducing	Small controlled addition.

Constituent	Typical role in green sand	Typical proportion / note
coal	carbonaceous layer near the mould-metal interface.	
Other additives	Wood flour, cereal, dextrin, iron oxide or special proprietary materials may be used to modify collapsibility, strength, expansion behaviour or surface quality.	Small percentage according to requirement.

Typical composition is not fixed: In examination answers, state the function of each constituent and, if percentages are given, identify them as typical ranges. Foundry sand is controlled by tests rather than by a single universal recipe.

8. Different Types of Sand Tests

Testing is essential because mould performance depends strongly on grain size, moisture, clay content, permeability, strength and compactability. Foundry sand tests are carried out on representative samples, often prepared using standard specimen dimensions and standard ramming energy.

Test	What it determines	Importance / interpretation
Moisture content test	Determines the percentage of water in the sand mixture. A weighed sample is dried to remove moisture and reweighed.	Moisture % = (loss in mass on drying / original wet mass) × 100. Moisture strongly affects green strength, flowability and permeability.
Clay content test	Measures the amount of clay or fine binder material associated with the sand.	The sample is washed/agitated so clay is removed in suspension; the dried residue is weighed. Excess clay usually reduces permeability and increases gas demand.
Grain fineness / sieve analysis	Determines grain-size distribution by passing dried sand through a standard set of sieves.	Used to calculate a grain-fineness number or characterize grain distribution. Fine grains improve surface finish; coarse grains improve permeability.
Permeability test	Measures the ease with which air passes through a standard compacted sand specimen.	A known volume of air is passed through the specimen under controlled pressure. Low permeability increases risk of gas defects.
Green compression strength	Measures compressive strength of a standard moist specimen.	Indicates ability of a green mould to retain shape and resist pressure/erosion during pouring.
Green shear strength	Measures resistance of moist sand to shearing action.	Useful for evaluating resistance to mould damage at edges, projections and parting surfaces.
Dry compression / dry strength test	Specimen is dried and its strength is measured.	Important for dry-sand moulds, cores and sand surfaces that become heated and dried during pouring.
Hot strength test	Strength is measured at elevated temperature.	Indicates ability to withstand thermal and metallostatic stresses while in contact with molten metal.
Mould hardness test	Measures surface hardness or resistance to penetration of a rammed mould.	Provides a rapid shop-floor indication of ramming/compaction consistency.
Compactability test	Measures the percentage decrease in height or volume of loose sand after standard compaction.	A practical control parameter in mechanized green-sand systems; it indicates how readily the sand will pack under moulding pressure.
Flowability test	Assesses the ability of sand to flow and distribute under ramming/squeezing.	Important for obtaining uniform mould density, especially around deep pockets and intricate patterns.
Shatter index / toughness-type test	Assesses the ability of compacted sand to resist breakage or crumbling under handling.	Used as an indicator of mould toughness and handling characteristics.
Loss on ignition (LOI)	Measures mass loss when sand is strongly heated.	Indicates the quantity of combustible/volatile constituents such as carbonaceous additives and organic material.
Acid demand / pH related tests	Used mainly where chemical binders are involved to assess sand chemistry and binder compatibility.	Especially relevant in chemically bonded sand systems.

Permeability concept: Permeability number is based on the volume of air passing through a standard specimen in a given time under a specified pressure. A commonly used form is **P = V H / (p A t)**, where V is air volume, H specimen height, p air pressure, A cross-sectional area and t time.

9. Binders

A binder is a material added to moulding or core sand so that individual sand grains adhere to one another and the mould or core develops sufficient strength.

9.1 Classification of binders

Class	Examples	Main characteristics / use
Clay / inorganic natural binders	Bentonite, fire clay, kaolinite-type clays	Common in green sand. Water activates the clay bond; economical and reusable.
Silicate binders	Sodium silicate (water glass)	Can be hardened rapidly by CO2 gas or ester processes. Produces strong moulds/cores.
Organic oil binders	Linseed oil and related drying oils	Traditionally used for cores; develop strength after baking.
Cereal / dextrin binders	Cereal flour, dextrin, starch-based materials	Improve dry strength, toughness and collapsibility in selected sand systems.
Synthetic resin binders	Phenolic, furan, urethane and other foundry resins	Provide high strength and dimensional accuracy in chemically bonded moulds and cores.
Cement-type binders	Portland cement or special refractory cements	Used for certain large moulds and specialized applications.

9.2 Characteristics of a good binder

- Provides adequate strength with a small addition.
- Distributes uniformly over sand grains during mixing.
- Does not reduce refractoriness excessively.
- Produces acceptable gas evolution during pouring.
- Allows sufficient permeability and collapsibility.
- Is economical, safe to handle and compatible with reclamation where required.

10. Additives

Additives are materials added in relatively small quantities to modify one or more properties of a moulding sand. They are not normally the main binder or base sand.

Additive	Purpose / effect
Sea coal / powdered coal	Produces a reducing carbonaceous atmosphere at the mould-metal interface; improves surface finish and helps prevent burn-on and metal penetration, especially in ferrous castings.
Wood flour / sawdust	Improves collapsibility and can help accommodate thermal expansion by burning out and creating voids.
Dextrin / cereal flour	Improves dry strength, toughness and mould handling characteristics.
Silica flour	Fine refractory filler; may improve surface finish in some mixtures but can reduce permeability if excessive.
Iron oxide	Can modify hot properties, reduce certain expansion-related defects and improve resistance to penetration in selected ferrous applications.
Graphite / carbonaceous material	Improves surface quality and resistance to metal penetration in selected moulds/cores.
Pitch / asphaltic materials	Traditionally used as carbonaceous additives to improve mould-metal interface behaviour.
Special proprietary additives	Used to control veining, expansion, penetration, gas generation, collapsibility or casting finish depending on alloy and process.

Selection principle: No additive is universally beneficial. The type and amount must be chosen to solve a specific moulding or casting problem without causing excessive gas, loss of strength or poor permeability.

11. Sand Preparation and Conditioning

Sand preparation converts raw or reclaimed sand, binder, water and additives into a uniform moulding mixture having the required compactability, strength and permeability. Good moulding results depend as much on correct preparation and control as on composition.

11.1 Main operations in sand preparation

1. **Receiving and storage:** New and reclaimed sands are stored separately where required. Moisture and contamination are controlled.
2. **Sand reclamation / shakeout recovery:** Used sand is separated from castings, lumps are broken down, tramp metal is removed and excessively degraded fines may be controlled or discarded.
3. **Screening / riddling:** Removes lumps, foreign matter and oversize particles.
4. **Proportioning:** Required quantities of return sand, new sand, clay/bentonite, water and additives are measured according to the sand-control recipe.
5. **Mulling / mixing:** The ingredients are mixed and worked in a sand muller. Mulling distributes binder and water as uniform films around the sand grains and develops the clay bond.
6. **Moisture adjustment:** Water is added in a controlled manner to achieve the target compactability and green strength. Uniform distribution is more important than simply adding a nominal percentage.
7. **Aeration / cooling:** Prepared sand may be aerated and cooled to improve flowability and avoid hot return sand causing unstable moisture and bonding conditions.
8. **Testing and correction:** Moisture, compactability, permeability, green strength and other properties are checked. Small corrections are made before the sand is released for moulding.
9. **Delivery to moulding station:** Conditioned sand is transported without excessive drying, segregation or contamination.

11.2 Conditioning of moulding sand

Conditioning means bringing the sand mixture to a consistent working state before moulding. It includes breaking lumps, cooling, restoring moisture, mixing in fresh binder/additives and aerating the sand so that it has uniform flowability and moulding properties.

- **Too dry:** Low plasticity and green strength; mould edges may crumble.
- **Too wet:** Low permeability, increased gas generation and possible sticking to the pattern.
- **Poorly mulled:** Uneven binder distribution gives local weak and hard spots.
- **Hot return sand:** Causes moisture evaporation and unstable moulding properties; cooling may be necessary.
- **Excess fines:** Reduce permeability and increase binder/water demand.

12. Sand Mould Making Procedure

The following is the standard hand-moulding procedure for a two-part green-sand mould using a split pattern. The exact sequence changes slightly for solid patterns, match-plate patterns, machine moulding or special gating arrangements.

Step	Operation	Explanation / purpose
1	Prepare the moulding equipment	Clean the pattern, flask, moulding board and tools. Check the pattern for damage and identify the parting surface, gating position and core requirements.
2	Place the drag	Place the drag flask upside down on the moulding board. Position the lower half of the split pattern with its parting surface on the board.
3	Apply parting material	Dust the pattern and board lightly with parting sand or suitable parting compound to prevent sticking.
4	Riddle facing sand	Sieve a layer of fine facing sand over and around the pattern, especially where

Step	Operation	Explanation / purpose
		good surface finish is required.
5	Fill the drag	Add backing/system sand in layers. Distribute the sand uniformly and avoid disturbing the pattern.
6	Ram the sand	Compact each layer using a hand rammer or suitable machine. Ramming should be firm and uniform: insufficient ramming gives weak moulds, whereas excessive ramming reduces permeability.
7	Strike off and vent	Level excess sand with a strike-off bar. Vent holes may be made with a vent wire to improve gas escape, taking care not to damage the pattern surface.
8	Turn the drag	Place a bottom board on the rammed drag and carefully turn the flask over so that the pattern parting surface is upward.
9	Fit the cope	Place the cope on the drag, aligning flask pins. Position the upper half of the pattern if a split pattern is used.
10	Locate sprue and riser pins	Place tapered sprue and riser pins at suitable positions. The sprue should lead to runners and ingates; the riser should feed regions likely to require liquid-metal compensation during solidification.
11	Apply parting sand	Dust the exposed drag surface and pattern parting surface so the cope sand does not stick to the drag.
12	Fill and ram the cope	Riddle facing sand where required, fill the cope with moulding sand, ram uniformly and strike off the top.
13	Vent the cope	Make vent holes where required, again avoiding the pattern cavity.
14	Remove sprue and riser pins	Withdraw the sprue/riser pins carefully and shape the pouring basin at the top of the sprue.
15	Separate cope and drag	Lift the cope vertically and place it safely aside. Examine the parting surface for broken edges or loose sand.
16	Withdraw the pattern	Loosen the pattern slightly by rapping and withdraw it vertically using draw screws or handles. Pattern withdrawal must avoid damaging the mould cavity.
17	Cut runner and ingates	Cut the runner and gates in the drag or at the parting line according to the gating design. Smooth sharp loose edges without overworking the mould.
18	Repair and finish the cavity	Repair minor damage using finishing tools. Remove loose sand. Apply mould wash/coating where required for surface finish or refractoriness.
19	Place cores, if required	Insert prepared cores into the core prints and support them properly. Chaplets may be used only where necessary.
20	Close the mould	Reassemble cope and drag using locating pins. Clamp or weight the mould sufficiently to resist metallostatic lifting force.
21	Final inspection before pouring	Check sprue, gates, vents, riser, core position, mould closure and cleanliness. The mould is then ready for pouring.

12.1 Common mould-making errors and their effects

Error	Likely consequence
Insufficient ramming	Weak mould, erosion, sand inclusions, distortion.
Excessive ramming	Low permeability and increased risk of gas defects.
Too much moisture	Gas defects, low permeability, sticking and poor mould stability.
Too little moisture / binder	Low green strength, crumbling edges and poor cavity definition.
Poor pattern withdrawal	Broken cavity edges and dimensional defects.
Loose sand left in cavity	Sand inclusions and rough casting surface.
Poor venting	Blows, pinholes and incomplete filling due to trapped gases.
Improper cope-drag alignment	Mismatch or mould shift.
Weak closing / inadequate weighting	Mould lift, run-out or excessive flash.

13.1 Property - meaning - defect prevented

Property	Meaning	If inadequate, possible problem
Refractoriness	Withstands molten-metal temperature	Fusion, burn-on, poor surface
Permeability	Allows gases to escape	Blows, pinholes
Green strength	Strength in moist state	Mould collapse, erosion
Dry / hot strength	Strength after drying / at high temperature	Cracking, erosion, mould failure
Flowability	Packs uniformly around pattern	Poor detail, density variation
Collapsibility	Breaks down after solidification	Hot tears, difficult shakeout
Fineness	Grain-size characteristic	Too coarse: rough surface; too fine: low permeability

Property	Meaning	If inadequate, possible problem
Adhesiveness	Sticks to flask walls	Mould may pull away or fall from flask

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